# CHITRANG POTDAR

Backend Engineer (Python) · FastAPI | Flask | AWS | Docker · AI/LLM Systems

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## PROFESSIONAL SUMMARY

Backend Engineer with production experience building Python-based APIs, cloud-native systems, and AI-integrated workflows. I take ownership end-to-end — from architecture and development to testing, CI/CD, and deployment. Currently deepening my focus on LLM-powered backend systems, having built serverless pipelines integrating AWS Transcribe and Claude API in production. I value reliability, clean API design, and systems that are built to last.

## TECHNICAL SKILLS

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| **Backend** | Python, Flask, FastAPI, RESTful API Design, JWT Authentication, 2FA (TOTP), Redis |
| **Databases** | PostgreSQL, Microsoft SQL Server — schema design, indexing, query optimisation |
| **DevOps & Cloud** | Docker, GitHub Actions, CI/CD Pipelines, AWS EC2, AWS Lambda, AWS S3, AWS Transcribe, AWS API Gateway |
| **Testing** | Pytest, Unit Testing, Coverage Reporting, Playwright |
| **AI/LLM** | Claude API, CrewAI, Gemini 2.0 Flash |
| **ML/Data** | Scikit-learn, XGBoost, NeuralProphet, Pandas, Numpy, MLflow, ZenML |
| **Languages** | Python (primary), SQL, JavaScript |
| **Tools** | Git, GitHub, Postman, Linux |

## EXPERIENCE

**Backend Engineer** · ArrayPointer Pvt. Ltd., Pune *Oct 2024 – Present*

***Investment Operations Platform***

***Stack:*** *Python, Flask, PostgreSQL, Redis, Docker, GitHub Actions, AWS EC2, Pytest, Playwright*

- Sole backend engineer for an Investment Operations Platform — owning the full backend lifecycle across feature development, debugging, testing, and production releases.
- Designed and implemented REST APIs for user management, transaction handling, and financial reporting — enforcing structured input validation, error handling, and consistent response contracts across all endpoints.
- Designed backend services with clear separation of concerns and structured validation to improve maintainability and reduce debugging overhead.
- Owned and implemented CI/CD pipelines and containerisation using GitHub Actions and Docker, ensuring consistent and reliable deployment workflows while eliminating manual release processes.
- Engineered Redis-based session management — auth tokens cached on login and served from Redis, eliminating redundant database queries on high-frequency authentication endpoints.
- Achieved 80%+ unit test coverage using Pytest, reducing regression defects across core API modules.
- Automated client-facing browser workflows using Playwright, saving ~2–3 hours/week of manual operational effort.

***Serverless Data Capture Pipeline***

***Stack:*** *AWS Lambda, AWS API Gateway, Google Sheets API*

- Designed and implemented a serverless data capture pipeline using AWS Lambda and API Gateway, routing frontend submissions into Google Sheets and eliminating the need for a dedicated database in this workflow.

***AI-Powered Meeting Intelligence Pipeline***

***Stack:*** *AWS Lambda, AWS API Gateway, AWS S3, AWS Transcribe, Claude (Anthropic API)*

- Architected an end-to-end serverless backend pipeline for meeting analysis — audio recordings and agenda documents are ingested from S3, transcribed via AWS Transcribe, and the resulting transcript is passed to Claude API alongside the agenda for structured analysis.
- Engineered the Claude API integration to produce structured outputs: a concise meeting summary, agenda adherence scoring (on-agenda vs. off-agenda coverage classification), and an auto-generated actionable task list — returned to the client via API Gateway.

**Python Developer Intern** · ArrayPointer Pvt. Ltd., Pune *Dec 2023 – Apr 2024*

***Financial Document Parsing Pipeline***

***Stack:*** *Python, PyPDF2, Camelot, Tabula, Pandas*

- Designed and developed an automated PDF parsing pipeline for financial documents (electricity bills, CAS statements), reducing manual extraction effort by ~70% and delivering clean, structured JSON output.
- Engineered a multi-strategy parsing layer combining PyPDF2, Camelot, and Tabula to handle diverse and inconsistent document layouts with consistent accuracy.

***Revenue Forecasting Model***

***Stack:*** *Python, XGBoost, Pandas, Scikit-learn*

- Developed a time-series revenue forecasting model using XGBoost with incremental learning — forecast accuracy improved by ~20% as new data was ingested, designed for continuous production use without full retraining.

## PROJECTS

TaskHub — Project Management Platform *Aug – Sep 2025*

**GitHub:** github.com/Dev-Chitrang/TaskHub-Full-Stack-Project

***Stack:*** *FastAPI, PostgreSQL, Docker*

- Designed and developed a scalable backend system using FastAPI and PostgreSQL supporting authentication, RBAC, workspace management, and task operations.
- Implemented JWT authentication with email verification, 2FA (TOTP), and password recovery; containerised with Docker using dev/prod config separation and structured API logging.

Retail DataOps & MLOps Platform *Mar – Jul 2025*

**GitHub:** github.com/Dev-Chitrang/data-warehouse-to-mlops

***Stack:*** *FastAPI, MSSQL, Scikit-learn, ZenML, MLflow, NeuralProphet, Docker*

- Engineered ETL pipelines ingesting ERP/CRM data into a structured SQL Server data warehouse (Bronze/Silver/Gold layers); exposed Demand Forecasting, Customer Segmentation, and Product Recommendation model outputs via FastAPI endpoints. Full stack containerised with Docker.
- Managed ML experiment tracking with MLflow and orchestrated model training and deployment pipelines with ZenML for reproducibility.

AI Trip Planner — Multi-Agent System *Sep – Oct 2025*

**GitHub:** github.com/Dev-Chitrang/Multi-Agent-Trip-Planner

***Stack:*** *CrewAI, Gemini 2.0 Flash, SerpAPI, Python*

- Designed and developed a multi-agent AI system using CrewAI — coordinating specialised Flight, Hotel, and Itinerary Planner agents powered by Gemini 2.0 Flash, with real-time data retrieval via SerpAPI.

## EDUCATION

**B.E. Computer Engineering** · Savitribai Phule Pune University, Pune *CGPA: 8.56 / 10* *Jun 2020 - Jun 2025*

## RESEARCH & ACHIEVEMENTS

- Published: Corporate Credit Rating Analysis using AI & Sentiment Analysis — IJIRCCE (Impact Factor: 8.65) Nov 2024
- Research Paper: github.com/Dev-Chitrang/Final-year-project/tree/Backend